

The empirical recurrence for OEIS A250591, recovered from its tail

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Abstract

OEIS A250591 records “Empirical recurrence of order 79” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 708$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 10$, so the recurrence is proved for every $n \geq 89$. Its last coefficient is $c_{79} = 60 \neq 0$, so no further tail satisfies anything shorter; any order-79 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A250591 is “Number of $(n+1) \times (7+1)$ 0..2 arrays with nondecreasing $\max(x(i,j), x(i,j-1))$ in the i direction and nondecreasing $\min(x(i,j), x(i-1,j))$ in the j direction.”. It has offset 1 and begins

110187, 855988, 5284985, 29029369, 145409040, 683598422, 3053953408, 13166219146, ...

The entry states:

Empirical recurrence of order 79 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 12:34:34, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 708$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 708$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 79: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 10$ is the first whose minimal order is exactly the stated 79.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 1416$ exact terms from $n = 10$ on, it returns order 79 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{79} c_i a(n-i),$$

c_1	28	c_{21}	−5865952837612	c_{41}	−12210763852312646	c_{61}	−9533305870657
c_2	−298	c_{22}	−1503274069035	c_{42}	−8149439297471104	c_{62}	−4537229603800
c_3	1070	c_{23}	26613440368903	c_{43}	17073443613441922	c_{63}	2956199644996
c_4	5870	c_{24}	−24999718126203	c_{44}	−1854800972418868	c_{64}	35325994254
c_5	−71061	c_{25}	−68765892055592	c_{45}	−13960399837095392	c_{65}	−451780122851
c_6	173557	c_{26}	157243672292997	c_{46}	8798160133904759	c_{66}	94507127062
c_7	954562	c_{27}	54712335994062	c_{47}	6113500590749698	c_{67}	33914341586
c_8	−6878018	c_{28}	−510810142678819	c_{48}	−9146459408235024	c_{68}	−16477442164
c_9	5658329	c_{29}	332688011018892	c_{49}	332743707522979	c_{69}	−76181879
c_{10}	86212536	c_{30}	985980151613436	c_{50}	5345418852786552	c_{70}	1320367397
c_{11}	−299502819	c_{31}	−1641096424242702	c_{51}	−2524240124830731	c_{71}	−207313452
c_{12}	−287010089	c_{32}	−778407388993214	c_{52}	−1675559871556999	c_{72}	−45574492
c_{13}	3749949176	c_{33}	3980327847871665	c_{53}	1872404874142175	c_{73}	16482363
c_{14}	−4984396497	c_{34}	−1633172002281566	c_{54}	−2579354458147	c_{74}	−304884
c_{15}	−22205694645	c_{35}	−5833035838668239	c_{55}	−746485246309549	c_{75}	−497322
c_{16}	77820896195	c_{36}	6963562000774028	c_{56}	269482899782198	c_{76}	62295
c_{17}	24952467961	c_{37}	4099155797905949	c_{57}	157085705792064	c_{77}	3149
c_{18}	−536404909197	c_{38}	−12832026893479697	c_{58}	−134224750748258	c_{78}	−1093
c_{19}	652891174221	c_{39}	2955753643054449	c_{59}	−1440272443510	c_{79}	60
c_{20}	1964385591893	c_{40}	14192644701344245	c_{60}	34412669218593		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 89$, and it is the recurrence OEIS A250591 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 10$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{79} - c_1x^{78} - \dots - c_{79}$. Its constant term is $-c_{79} = -60$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 79 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 79, so $\deg q = 79$, and it is monic. It holds from some index t on. From $\max(t, 10)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 79, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 79, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 60.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-79 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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